Md Kawsher Mahbub

Department of Computer Science and Engineering
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Google Scholar: scholar.google.com/citations?user=fQ_TTFsAAAAJ

RESEARCH INTERESTS

Artificial Intelligence, Deep Learning/Machine Learning, Health Informatics, Computer Vision, Medical Image Analysis, Uncertainty Quantification, Out-of-Distribution Generalization, Explainable AI

EDUCATION

MS in Computer Science Aug. 2023 – May 2025
Kent State University, Kent, OH, USA
CGPA: 3.80

Bachelor of Science in Computer Science and Engineering Feb. 2017 – May 2021
Bangladesh University of Business and Technology (BUBT), Dhaka, Bangladesh
CGPA: 3.84

PROFESSIONAL EXPERIENCE

Assistant Professor Mar. 2026 – Present
Department of Computer Science and Engineering
NPI University of Bangladesh, Singair, Manikganj

Graduate Teaching Assistant Aug. 2023 – Dec. 2025
Department of Computer Science
Kent State University, Kent, OH, USA

Lecturer Dec. 2021 – Jul. 2023
Department of Computer Science and Engineering
Bangladesh University of Business and Technology (BUBT), Dhaka, Bangladesh

PUBLICATIONS

Under Review:

UR2. **M. K. Mahbub**, S. Hossain, M. A. M. Miah, M. N. Morshed, and M. Biswas, Beyond Benchmark Accuracy: Forensic Explainability Auditing for Ensemble Chest Radiograph Classifiers, submitted to Medical Image Analysis (MedIA).

UR1. **M. K. Mahbub**, M. Biswas, M. N. Morshed, and W. Yu, SpatialUQ: Post-Hoc Uncertainty Quantification from Spatial Consistency in Black-Box Vision Models, submitted to NeurIPS 2026.

Published Journal Papers:

J3. Ghose, P., Oliullah, K., **Mahbub, M. K.**, Biswas, M., Uddin, K. N., & Jamil, H. M. (2025). Explainable AI assisted heart disease diagnosis through effective feature engineering and stacked ensemble learning. Expert Systems with Applications, 265, 125928.

J2. **Mahbub, M. K.**, Biswas, M., Gaur, L., Alenezi, F., & Santosh, K. C. (2022). Deep features to detect pulmonary abnormalities in chest X-rays due to infectious diseases: COVID-19, pneumonia, and tuberculosis. Information Sciences, 592, 389-401.

J1. Ghose, P., Alavi, M., Tabassum, M., Ashraf Uddin, M., Biswas, M., **Mahbub, M. K.**, & Zhao, Z. (2022). Detecting COVID-19 infection status from chest X-ray and CT scan via single transfer learning-driven approach. Frontiers in Genetics, 13, 980338.

Published Conference Papers:

C7. **Mahbub, M. K.**, Zamil, M. Z. H., Miah, M. A. M., Ghose, P., Biswas, M., & Santosh, K. C. (2022, July). Mobapp4infectiousdisease: Classify COVID-19, pneumonia, and tuberculosis. In 2022 IEEE 35th International Symposium on Computer-Based Medical Systems (CBMS) (pp. 119-124). IEEE.

C6. **Mahbub, M. K.**, Biswas, M., Miah, M. A. M., & Kaiser, M. S. (2022, February). Deep neural networks for brain tumor detection from MRI images. In Proceedings of the Third International Conference on Trends in Computational and Cognitive Engineering: TCCE 2021 (pp. 473-485). Singapore: Springer Nature Singapore.

C5. **Mahbub, M. K.**, Miah, M. A. M., Islam, S. M. S., Sorna, S., Hossain, S., & Biswas, M. (2021, November). Best eleven forecast for Bangladesh cricket team with machine learning techniques. In 2021 5th International Conference on Electrical Engineering and Information Communication Technology (ICEEICT) (pp. 1-6). IEEE.

C4. **Mahbub, M. K.**, Biswas, M., Miah, A. M., Shahabaz, A., & Kaiser, M. S. (2021, July). COVID-19 detection using chest X-ray images with a RegNet structured deep learning model. In International Conference on Applied Intelligence and Informatics (pp. 358-370). Cham: Springer International Publishing.

C3. Biswas, M., **Mahbub, M. K.**, & Miah, M. A. M. (2021, December). An enhanced deep convolution neural network model to diagnose Alzheimer's disease using brain magnetic resonance imaging. In International Conference on Recent Trends in Image Processing and Pattern Recognition (pp. 42-52). Cham: Springer International Publishing.

C2. Biswas, M., Nova, A. J., **Mahbub, M. K.**, Chaki, S., Ahmed, S., & Islam, M. A. (2021, August). Stock market prediction: A survey and evaluation. In 2021 International Conference on Science & Contemporary Technologies (ICSCT) (pp. 1-6). IEEE.

C1. Biswas, M., Niamat Ullah Akhund, T. M., **Mahbub, M. K.**, Saiful Islam, S. M., Sorna, S., & Shamim Kaiser, M. (2021). A survey on predicting player's performance and team recommendation in game of cricket using machine learning. In Information and Communication Technology for Competitive Strategies (ICTCS 2020) ICT: Applications and Social Interfaces (pp. 223-230). Singapore: Springer Singapore.

STANDARDIZED TEST SCORES

Graduate Record Examination (GRE) 23rd September, 2021

Total	Quantitative	Verbal	Analytical
303	154	149	2.5

SERVICES

Program Committee Member

- IEEE CBMS International Symposium on Computer-Based Medical Systems, 2023–2026
- The 35th International Conference on Artificial Neural Networks (ICANN 2026)

Reviewer for Journal Manuscript Submissions

- CMC – Computers, Materials & Continua (Tech Science Press)
- Expert Systems (Wiley)
- International Journal of Intelligent Transportation Systems Research (Springer)

TECHNICAL SKILLS

Programming Languages	Python, C, C++, C#, Java
Web Technologies	HTML5, CSS3, PHP, WordPress
Frameworks & Libraries	PyTorch, TensorFlow, OpenCV, NumPy, SciPy, Scikit-learn, Pandas, Keras, Matplotlib
Databases	MySQL, Oracle, MongoDB
Development Tools	Jupyter Notebook, Anaconda, PyCharm, Visual Studio, Net-Beans, Eclipse, Android Studio
Miscellaneous	Git, LaTeX, UML, E-R Diagram

ACADEMIC PROJECTS

1. Crowd Source Recipe Collections (Master's Capstone Project)
2. Deep Neural Network for Pulmonary Abnormality Screening Using Chest X-ray: COVID-19, Pneumonia, and Tuberculosis
3. Infection Status from Chest X-ray and CT Scan via Single Transfer Learning Driven Approach
4. Hostel Management System

AWARDS AND HONORS

First Place and People's Choice Award, International Cook-Off 2024, Kent State University — Sep. 2024

Merit-based Scholarship (5 times), Bangladesh University of Business and Technology (BUBT)

LEADERSHIP

Secretary — Nov. 2024 – Sep. 2025
Bangladesh Student Association at Kent State University (BSA-Kent)

Advisor — Oct. 2022 – Jul. 2023
IEEE CS BUBT Student Branch Chapter

Member, Registration & Web Site Committee — Jul. 2022 – Oct. 2022
ICPC Asia Dhaka Regional Contest 2021

VOLUNTEERING / EXTRA-CURRICULAR ACTIVITIES

- Got 120+ accepted solutions at URI Online Judge (urionlinejudge.com.br · profile/193022)
- Got 50+ online judge accepted solutions (stopstalk.com/user/profile/kawsher11)
- Participated in intra-university programming contests

PROFESSIONAL MEMBERSHIP

Member, IEEE

Deep Learning Research Group (DLRG), Department of Computer Science and Engineering, Bangladesh University of Business and Technology (BUBT)

REFERENCES

Milon Biswas (Undergraduate Thesis Supervisor)
Ph.D. Candidate & Adjunct Faculty
Towson University, Towson, Maryland, USA
Email: mbiswas1@students.towson.edu